
Causal Steering of Protein Language Models: When Correlational Validation Fails, and How to Catch It

Dmitriy Ivkov
University of Michigan
divkov@umich.edu

Abstract

1 Activation steering has recently been shown to reproduce a known protein property
2 (thermostability) in ESM2-650M via difference-of-means vectors. We ask whether
3 a scoring proxy’s correlation with real experimental labels predicts whether steering
4 toward it actually works, and find the answer is no. Under a pre-registered
5 six-criterion protocol testing five novel target properties, our best-validated proxy
6 (binding affinity, $r = 0.80$ held-out) steers nothing, while a substantially weaker
7 one ($r = 0.22$, catalytic activity) produces a robust, three-seed-replicated effect.
8 This mirrors an independent, component-level finding in the same model family:
9 attention heads ranked by correlation with real 3D contacts show near-zero relation-
10 ship to their causal ablation effect. We show that a single-seed steering result
11 can mislead in criterion-specific ways. For one target, the effect direction replicates
12 across three seeds, but the residue-exclusion check passes on only two. We also
13 introduce a lightweight diagnostic based on steering-vector cosine similarity across
14 targets. It explains an otherwise unexplained ambiguous result as a geometric echo
15 of a different property’s validated direction. We release the full protocol, all target
16 harnesses, and the seed-replication data.

17 1 Introduction

18 Difference-of-means activation steering adds a direction to a model’s residual stream. The direction
19 is built from the mean activation difference between high- and low-property example groups. This
20 method reproduces known thermostability biology in ESM2-650M [Huang et al., 2025]. We ask the
21 natural follow-up: does a scoring proxy’s correlation with real labels predict whether steering toward
22 a new property works? Across five novel targets under one pre-registered protocol, the best-validated
23 proxy in our study ($r = 0.80$, binding affinity) produces a flat null effect, while a substantially weaker
24 proxy ($r = 0.22$, catalytic activity) produces a clean, three-seed-replicated effect. This mirrors an
25 independent, component-level result: Vig et al. [Vig et al., 2021] found an attention head whose
26 correlation with real 3D contacts is $12.9\times$ enrichment, explicitly flagged as untested causally; ablating
27 it gives an effect indistinguishable from a low-correlation control, and a full 480-head sweep ranks
28 the correlational top pick 313th of 480 by actual causal effect. Two levels, two techniques, one lesson:
29 correlational signal is evidence about representations, not about intervention. Our contribution is the
30 audit discipline needed to trust a causal claim here, demonstrated via two near-misses caught in our
31 own pipeline before they could be reported as clean results.

32 2 Method

33 We lock six criteria before testing any target and compute them automatically. Criterion 1 requires
34 the real direction to beat a matched-norm random control via paired bootstrap with 10,000 resamples.

Table 1: Five novel targets. Proxy r : Pearson correlation vs. real labels, held-out, before any steering run.

Target	Proxy r	Steering result	Verdict
Binding affinity	0.80	flat null, all α	KILL
Catalytic activity	0.22	PASS, 3/3 seeds	PASS
Intrinsic disorder	0.45	3/3 direction; 2/3 robustness	PASS*
Immunogenicity	≤ 0.10	not run	KILL (pre-run)
Expression yield	0.31	fails residue-exclusion	AMBIGUOUS

*PASS on criteria 1,2,4,6, and 2/3 seeds on criterion 3.

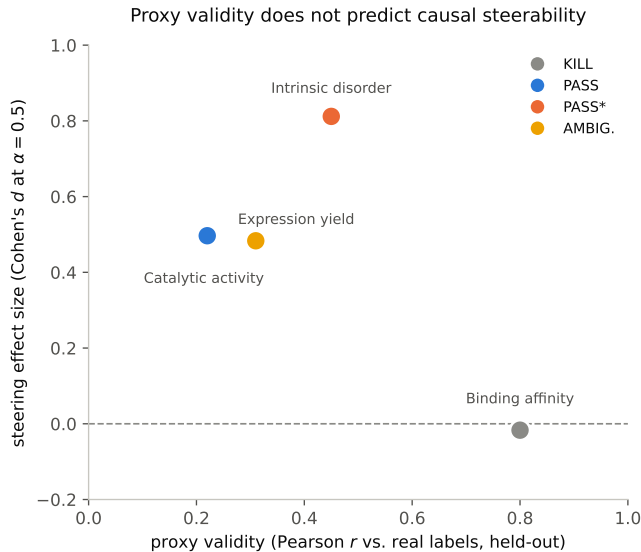


Figure 1: Proxy validity vs. steering effect size (Cohen’s d , $\alpha = 0.5$). The best-validated proxy (binding affinity, $r = 0.80$) gives the smallest effect; a proxy less than half as strong (catalytic activity, $r = 0.22$) gives one of the two largest. No positive trend.

35 Criterion 2 requires a monotonic dose-response across at least three strengths within a safe range
36 (§4). Criterion 3 requires the effect to survive residue-exclusion, and criterion 4 requires the proxy to
37 be validated against real labels before the run. Criterion 5 requires the result to beat the best existing
38 technique when one exists, and criterion 6 requires at least 150 held-out sequences. A target failing
39 any criterion is a KILL; one passing 1–2 but failing 3 or 4 is AMBIGUOUS.

40 3 The Central Finding: Proxy Validity Does Not Predict Causal Effect

41 **Binding affinity.** We use ProteinGym DMS data for KRAS binding to a DARPIn. Its mutational-
42 sensitivity-weighted proxy is validated more rigorously than any other target here: $r = 0.80$ held-out,
43 a weight-shuffle null of $r = -0.001$, held-out-position generalization of $r = 0.54$, and single-to-
44 double-mutant extrapolation of $r = 0.70$. Yet the steering effect is a flat null at every α , with zero
45 degenerate sequences at any strength. A plausible mechanism is that the vector-building groups are
46 single-backbone point mutants, only 1–2 residues from a shared reference. Difference-of-means
47 therefore has little compositional signal to use. The mean compositional distance between low and
48 high groups is 0.003, compared with 0.02–0.05 for the three cross-protein targets below.

49 **Catalytic activity.** On the cross-protein DLKcat dataset, a glycine-minus-arginine proxy matches
50 the published activity-stability tradeoff in enzymology. It is the weakest proxy among the four
51 runnable targets, with $r = 0.22$, but produces a clean, monotonic, residue-robust effect (Figure 2,

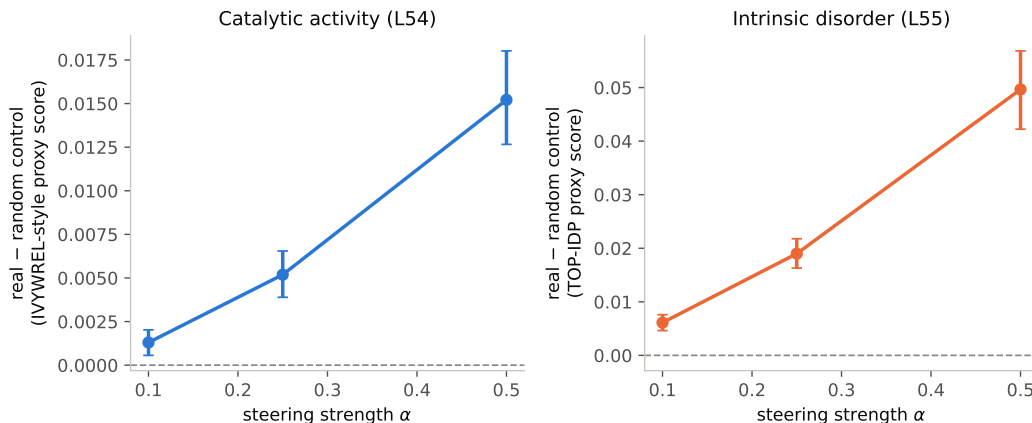


Figure 2: Dose-response, the two targets passing criterion 2. Both monotonic, 95% bootstrap CIs bounded away from zero throughout.

left) replicated identically across three independent seeds. It is the only target with fully unanimous seed robustness.

Intrinsic disorder. The DisProt proxy uses the TOP-IDP scale. It is the strongest proxy here, with $r = 0.45$ and a per-residue AUC of 0.71. The effect is significant and monotonic on all 3 seeds (Figure 2, right), but its residue-exclusion check passes on only 2. Those seeds retain 36–42% of the magnitude. On the third seed, 10% is retained and the confidence interval crosses zero. Existence and direction are seed-stable, but the artifact check is not.

Immunogenicity. Across four IEDB tiers, proxies predictive on a surrogate binding-affinity tier ($r = 0.37$ – 0.43) collapse to near chance on the real T-cell-response endpoint ($r \leq 0.10$). A full-length-antigen proxy reaches $r = 0.38$, but organism identity explains 58% of the apparent signal. Its correlation flips to $r = -0.32$ once organism is held out of cross-validation. The target is killed before any steering run.

Expression yield. The eSol net-charge-deviation proxy validates at $r = 0.31$ – 0.34 . Its effect is significant and dose-responsive but collapses under residue-exclusion. Rather than discarding this result, we compute steering-vector cosine similarity against every other target: $+0.38$ overall with intrinsic disorder’s vector, rising to $+0.56$ – 0.67 at the three deepest layers. This is consistent with literature showing that disordered regions reduce solubility and expression. The result is best read as a partial echo of disorder’s already-validated effect through a more collapse-prone proxy, not a sixth steerable property.

4 Self-Correction Case Studies

A spurious pass from an unsafe comparison. Testing whether restricting steering to the 5 layers previously found causally necessary preserves the full 33-layer effect, a first pass selected its best-performing α from the whole tested range, including $\alpha \geq 1.0$, and reported a clean pass at $\alpha = 2.0$. This was comparison artifact. At that α , the 33-layer arm had already fully collapsed into degenerate output, leaving zero non-degenerate sequences. The 5-layer arm had not collapsed, so it won only by breaking down later. After restricting the comparison to a range where neither arm has collapsed and rerunning end to end, the 5-layer subset retains only 43–59% of the full effect.

A criterion passing on only 2 of 3 seeds. The disorder target’s residue-exclusion check (§3) is seed-sensitive even though the headline effect is not. A single seed could have reported either a unanimous pass or a false kill on this criterion. Only a multi-seed rerun reveals the instability.

5 Related Work

Huang et al. [2025] validate this steering method on thermostability; we extend it to five new properties under a pre-registered protocol and test, to our knowledge for the first time, whether proxy correlation strength predicts steering success across unrelated properties. Vig et al. [2021] identify the head we redo as a genuine causal test. Adams et al. [2025] train SAEs on ESM-2 and show through correlational linear probing that thermostability and localization determinants are identifiable in SAE feature space. Their one steering check is a coherence test, not validation against a real label, and their discussion states “investigating how to steer the model in useful ways remains an open direction for future work.” We answer a version of that for a non-SAE mechanism, specifically for whether correlational validity predicts causal usefulness.

6 Discussion

Every steering result here is causal about the model’s activation space. It reflects a real intervention evaluated against a random-direction control and a residue-exclusion check, but it is not validated against a wet-lab assay. A PASS means “causally steers a validated proxy,” not “causally engineers the biological property.” The vector-geometry check that explained the expression-yield result requires only one extra embedding pass. It is worth running by default whenever multiple targets are tested together.

All experiments use one model, ESM2-650M, at one scale, and all proxies are compositional heuristics. Each target’s proxy was selected as the best of several candidates on the same held-out split used for its reported r , so the values should be read as upper bounds. We ran many significance tests without a multiple-comparisons correction, although requiring multi-alpha and multi-seed agreement for a PASS partly mitigates this issue. Seed robustness was checked for only two of five targets, and generated sequences received no structural or functional plausibility check.

7 Conclusion

Correlational validity does not predict causal effect when tested at two levels of the same model class. This applies both to an attention head’s biological alignment and to a proxy’s correlation with real labels. Proxy validation strength is not a reliable filter for which targets are worth attempting, and a single steering run should not be trusted without a seed-robustness check where the result matters downstream.

Responsible-use statement

This work identifies methodological pitfalls in causally validating activation-steering claims on protein language models and does not introduce a new capability for generating harmful biological sequences; all target properties studied (thermostability, binding, catalysis, disorder, immunogenicity, expression) are common protein-engineering objectives with dual-use risk no greater than the underlying published steering method [Huang et al., 2025] we extend. We see no additional mitigation specific to this paper’s contribution beyond standard biosecurity practice already governing protein-language-model research.

References

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